KI Service Zentrum by Hasso-Plattner-Institut

HPI

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Bundesministerium für Forschung, Technologie und Raumfahrt

Workshop
How to use our HPC Cluster

Felix Boelter and David Goll



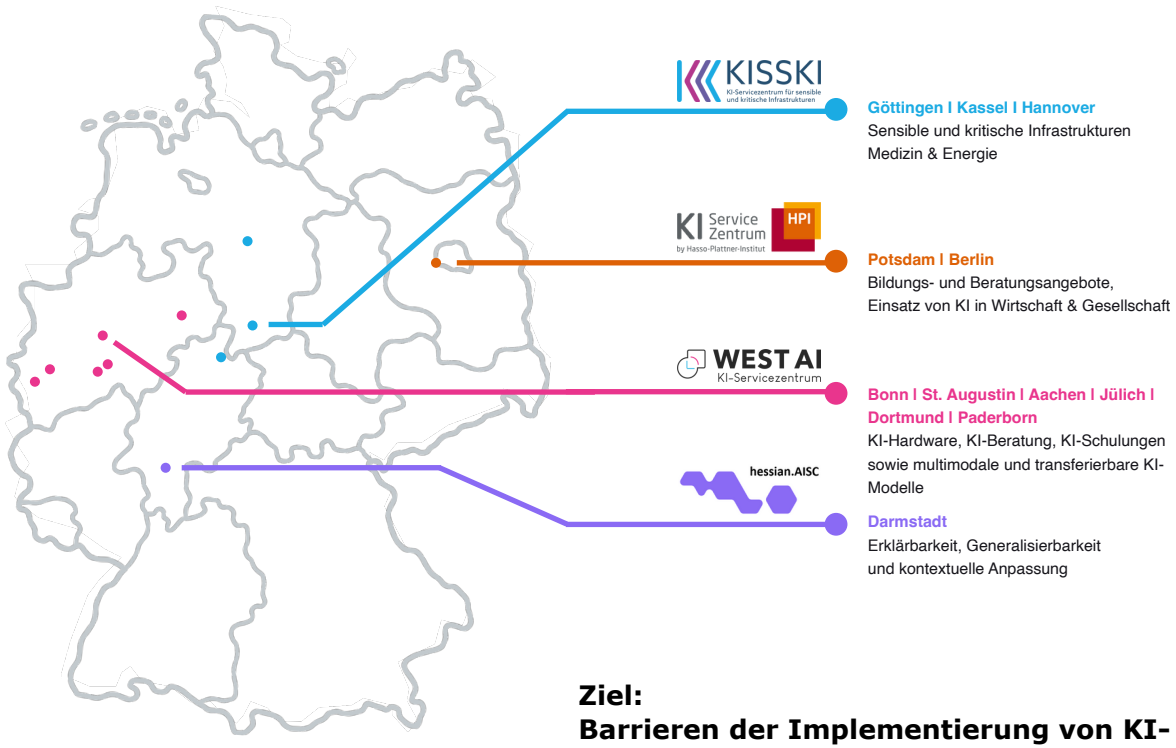
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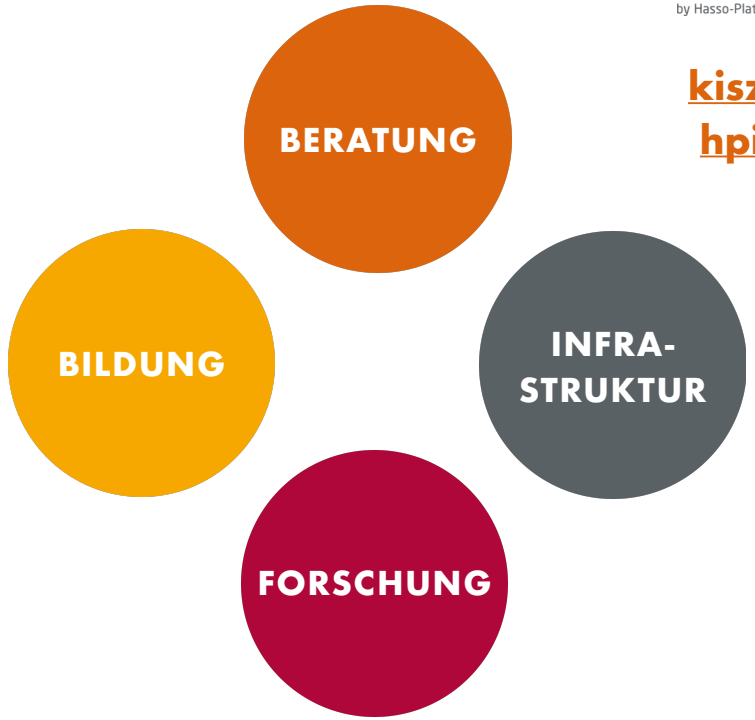
KI-Servicezentrum Berlin-Brandenburg

KI Service Zentrum by Hasso-Plattner-Institut HPI

kisz@hpi.de
hpi.de/kisz



Ziel:
Barrieren der Implementierung von KI-Anwendungen in Gesellschaft und Wirtschaft reduzieren



Hasso-Plattner-Institut gGmbH

- Bildet mit **Universität Potsdam** die **Digital Engineering Fakultät**
- Vereint **Forschung, Lehre** mit den Vorteilen einer **privat finanzierten, gebührenfreien Institution**
- Besteht aus Einrichtungen wie der **E-School**, der **D-School,**
- dem **Mittelstandsdigitalzentrum** und **dem KI-Servicezentrum**

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Newsletter

BILDUNG

Talks

tele-task.de/series/1463

- Gastvorträge zu Forschung und Innovation





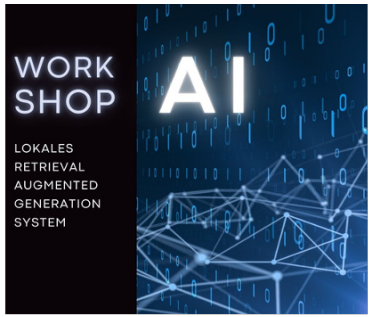


Work shops

aimaker.community

- Praxisnahe Themen
- Beispielthemen: Speech2summary, Docker für ML, semantische Suche





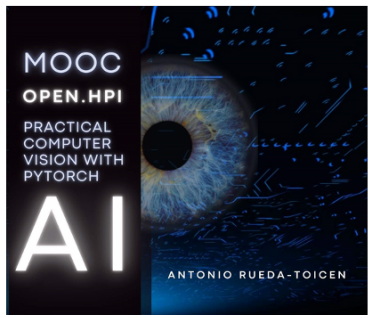


MOOCs

open.hpi.de/channels/ai-service-center

- ChatGPT: Was bedeutet generative KI für unsere Gesellschaft?
- Profitable KI
- KI Biases verstehen und vermeiden







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Sprechstunde buchen

BERATUNG

KI-Sprechstunde

- Beantwortung von Fragen:
  - zu KI-Infrastruktur
  - zu KI-Modellen & Frameworks

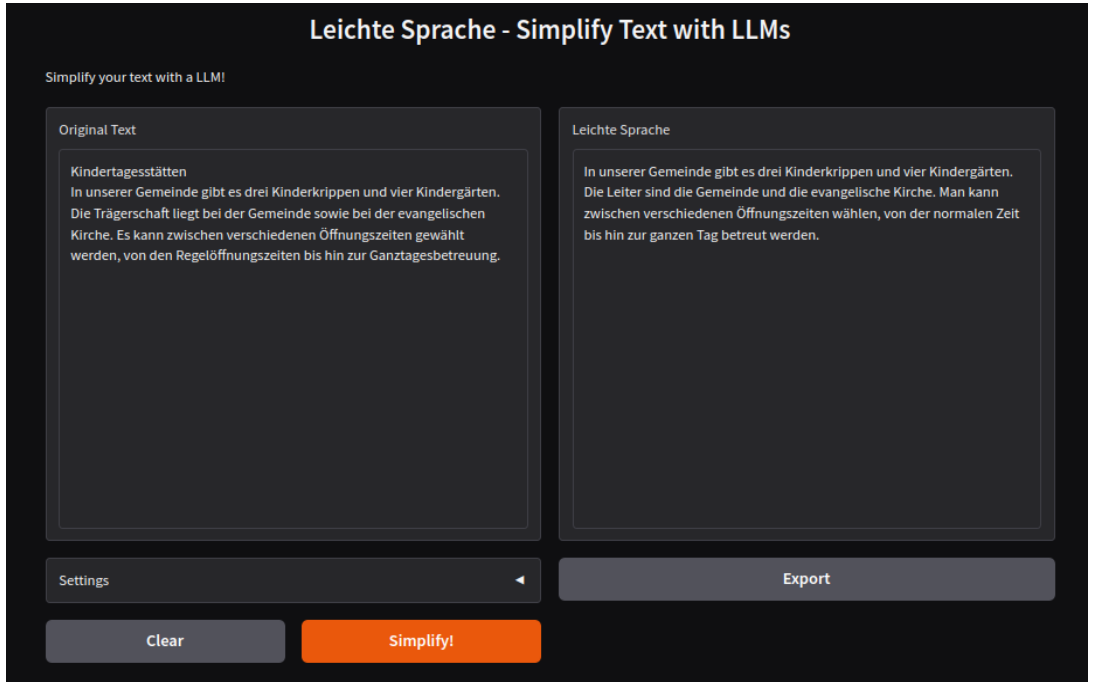
KI-Pilotprojekte

- Co-Entwicklung eines Prototyps
- Bewerbung alle drei Monate
- Auswahlkriterien z.B. KI-Reife, Gemeinwohl
- Veröffentlichung der Ergebnisse

Kooperationen

- Gemeinsam organisierte Netzwerktreffen





github.com/aihpi/leichte-sprache

Bisherige KI-Pilotprojekte

- Generierung Mathematik-Problemen
- Leichte Sprache
- Generierung von Upcycling Vorschlägen
- Reduzierung von Food Waste
- Datierung mittels Handschrift

Jetzt bewerben!

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Zugangsanfrage
aisc.hpi.de

INFRASTRUKTUR





- Zugang **kostenfrei**
- Kein Produktionsbetrieb
  - Daten sollten **anonymisiert** oder **synthetisiert** sein
  - kein **Hosting** von Produkten
- **Reporting** & **Veröffentlichung** durch Nutzende
- **Altrechte** bleiben bei Nutzenden
- **Neurechte** bleiben bei Nutzenden
  - Einräumen von Nutzungsrechten für Forschung und Lehre

Training
- 64 NVIDIA H100 GPU

Inferenz
- 40 NVIDIA A30 GPU

ARM Server
- Ampere Altra Max M128-30 CPU
- 2 x NVIDIA L40 GPUs

GPU Server
- AMD Epyc CPU
- 8 x NVIDIA L40S GPU

Edge
- ARMv8 CPU
- NVIDIA Jetson AGX Module

Neuromorph
- 288 SpiNNaker2 Chips

Speicher
- 1.5 PB NVRAM

Netwerk
- 400 Gb/s Infiniband
- 200 Gb/s Ethernet

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AGENDA

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Workshop Structure

Part	Topic	Duration
I	Introduction and Motivation	10 min
I	Interactive: Cluster access setup	20 min
I	Slides: partitions, nodes, SLURM, Interactive vs. Batch	10 min
I	Interactive: VSCode Installation	5 min
	Break	
II	Interactive SLURM workflow	10-20 min
II	Slides: Basics of CPUs/GPUs and fairshare	10-20 min
II	Interactive: Cloning of repository	5 min
II	Interactive: Batch scripts (`01`-`08`)	20 min
II	Conclusion and Feedback	5 min

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Motivation

Scale: Train models that don’t fit on consumer hardware

Speed: Powerful GPUs, run experiments in parallel

Freedom: Submit and disconnect

Collaboration: Shared environment across your team

Cost: Free for AISC projects





Documentation

Please make yourself familiar with the HPC documentation!

If you have questions or need assistance, consult our FAQ or reach out to us:

For technical issues:
aisc-helpdesk@hpi.de

For organizational issues:
ki-servicezentrum@hpi.de

Cluster (HPC/Slurm)

SLURM (Scheduler)
Slurm Basics
Job Examples
Our Customizations
Scheduling and Priorities
Environmental Variables
Slurm REST API

Processes
Access to the HPC Cluster
Group Shares / Project Storage
Password Reset

Resources
Partitions
Hardware
Naming Convention
Node Features
Login Nodes
Run Nodes

Containerization
Enroot
Pyxis (Slurm Container Plugin)
Using Slurm within a container

Storage / Data Transfer
Storage Overview
Quotas / Limits
Scratch Space
Data Transfer

Software Installation
Python / Conda
Sudo / Superuser Permissions
Visual Studio Code
Jupyter Lab / Notebook
Matlab
PyCharm
R Language / RStudio

Cluster (HPC/Slurm)

The cluster offers powerful shared resources (CPU and GPU) for your computation jobs. You can run them interactively (in a shell like on your computer) or spawn a bunch of them at once with a batch file.

See Slurm Basics for the general concepts and more details on job management.

Tip

No access to the HPC cluster yet? Please check the access documentation first.

Quick Start

1. Request access via sc-helpdesk@hpi.de (if you don't have it yet)
2. Join the Slack channel #sc-announce for general announcements (e.g. maintenance, etc.), as well as #sc-discuss for questions and discussion about the cluster with other users.
3. Connect to the Scientific Compute VPN (This is not the same as the HPI VPN!)
4. Connect via SSH (name must be lowercase): `ssh {first.lastname}@hpc.sci.hpi.de`
5. Pick a partition: e.g. `cpu-interactive` (also see Partitions)
6. Pick a account you are allowed to use: e.g. `sci-demo-account` (run `saccount` to see your accounts)
7. Run the job, e.g. `srun --partition=cpu-interactive --account=sci-demo-account --cpus-per-task=2 --gpus=0 --mem=1G --time=00:30:00 --pty bash`
8. Run your job: `python3 myscript.py` (also see Job Examples)
9. Leave the node: `exit` or `^ Ctrl` + `D`

Attention

Do not run anything on the access nodes, but on the compute nodes allocated to you via Slurm. The access nodes block long running jobs and may be restarted monthly.

Do not run vscode on the login nodes! Please see our guide to using vscode on the cluster or use a Run Node instead.

Tip

See the Software Installation section in the menu on the left for instructions on how to install common data science tools like R, Jupyter Notebooks, or Conda.

Table of contents
Quick Start
SSH Public Key Authentication
SSH Host Key Verification

Initial Setup

https://github.com/aihpi/workshop-slurm

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Getting Started

Cluster Access Setup

You can find all of this information in the Scientific Compute Documentation. Please note that the Docs provide information for the general HPI HPC, of which the AISC infrastructure is a part of. We outline the important steps below:

1. Get access via the AISC portal by submitting a proposal:
2. When your account is created, you should receive a .zip file via mail, containing your username, initial password, and, crucially, an .ovpn file.
3. Set your password via the steps explained in the .zip file
4. Set up a VPN connection and keep it running
5. Set up a SSH connection with the cluster by following the steps outlined to the left in SSH guide
6. With your VPN connection **active**, connect to the cluster via SSH: `ssh user.name@hpc.sci.hpi.de`

> ***Optional:*** Instead of typing the full address every time, create or edit `~/.ssh/config` and add:
>
> ```
> Host hpi-hpc
>     HostName hpc.sci.hpi.de
>     User user.name
> ```
>
> After saving, you can connect with just `ssh hpi-hpc`. You can replace `hpi-hpc` with any alias you prefer.

7. Once you are connected, please take some time to familiarise yourself with the basic concepts of SLURM, jobs (especially SBatch files), partitions (only the aisc-... partitions are relevant for you), and run nodes, `nvidia-smi`, ...

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Cluster Basics



Nodes

Cluster Basics

- Cluster = many computers (nodes) connected over a network
- Each node has its own CPUs, RAM, (sometimes) GPUs, …
- Three types of nodes on our cluster
    1. Login node (lx01)
        - Entry point, like a lobby
        - Used only for file management, submitting jobs, …
    2. Run node (rx01/rx02)
        - Lightweight interactive work, VSCode, …
        - Shared usage. Per-user limit: 4 CPUs, 8GB RAM
    3. Compute node (gx[07-14], gx15v1, gx17v1, …)
        - Any computation (also interactive), ML training, data processing, …
        - Exclusive usage. Via SLURM only!

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SLURM Basics

What is SLURM?

- Job scheduler: manages who gets what resources and when
- You request resources through SLURM
- Jobs are prioritised based on **availability** and **fairness**

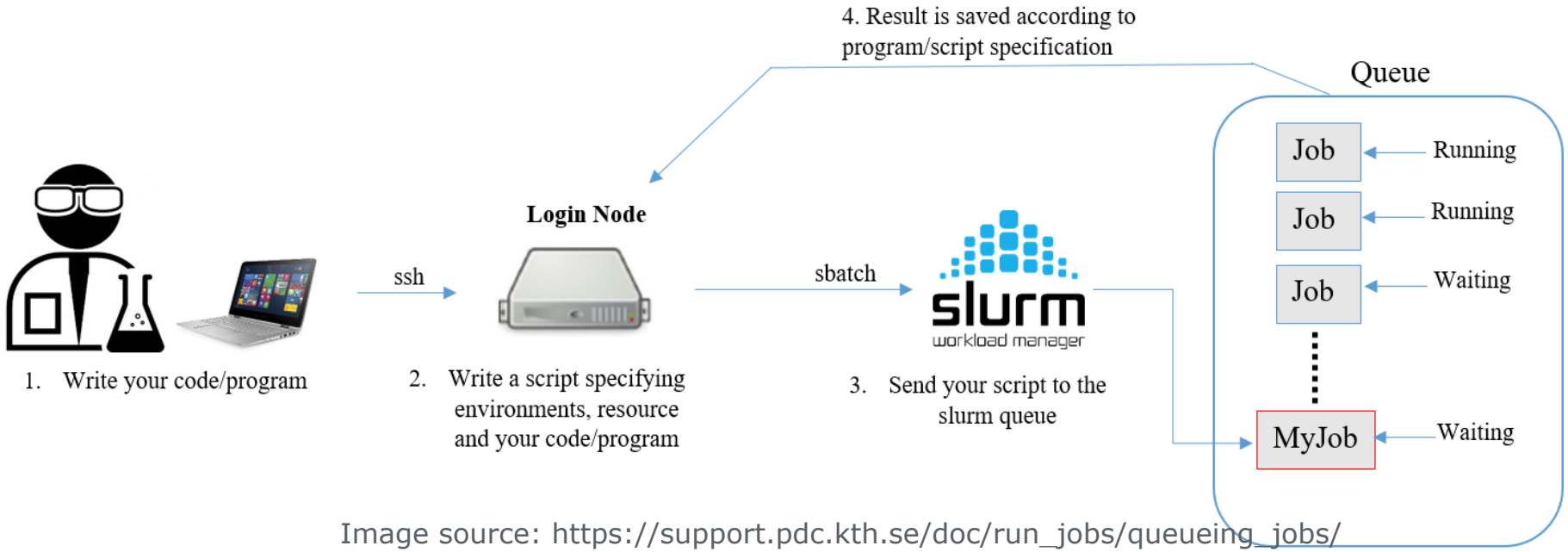


Image source: https://support.pdc.kth.se/doc/run_jobs/queueing_jobs/

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Key Concepts

- Partition: group of nodes with shared rules (time limits, hardware type)

Partition	Time Limit	Use Case
pot-hpi-aisc-shortrun	1h	Quick interactive tests (no sbatch)
pot-hpi-aisc-interactive	8h	Interactive terminal, testing
pot-hpi-aisc-batch	7 days	Long training runs (only accessible via sbatch jobs)

- Account: Your project's billing code. In our case it is always `aisc`.

  —> You can find out which accounts you have via `saccount`

- Quality of Service (QOS): Used to prioritize certain jobs. We use `aisc`.

  —> You will likely not need to use this

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SLURM Commands

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Command	Functionality
sinfo	Overview of the CPU and GPU usage of the cluster (wrapped for prettier output)
sfree	View available resources to select them for a job
squeue	View submitted jobs and their state
sjob	Show details about a certain job
snode	Show details about a certain node
salloc	Obtain a Slurm job allocation (a set of nodes), execute a command, and then release the allocation when the command is finished
srun	Execute command on the resources assigned to a job
saccount	Show the accounts you are allowed to use
sbatch	Submit a batch script to Slurm
swatch	View current resource usage for the active job

https://docs.sc.hpi.de/cluster/SLURM/Basics/

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Two ways to work on compute nodes

	Interactive		Batch
Command	srun	salloc/sblock	sbatch
Output	Live in terminal	Live in terminal	Written to log file
Best for	Quick commands, sanity checks	Interactive work	Training runs, anything serious

```
[d.goll@lx01:~$ cat test.py
print("Hello World!")
[d.goll@lx01:~$ python test.py
This command is not allowed on the login node!
Please run resource-intensive commands in a Slurm job instead to avoid overloading the login node.
HINT: Use 'srun' to start an interactive job on a compute node.
HINT: Use a run node for lightweight interactive tasks.
      See https://docs.sc.hpi.de/cluster/Resources/Run-Nodes/
[d.goll@lx01:~$ srun --account=aisc --partition=aisc-interactive --time=00:01:00 --mem=1G --pty bash
[d.goll@gx15v1:~$ python test.py
Hello World!
[d.goll@gx15v1:~$ exit
exit
```

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Two ways to work on compute nodes

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[d.goll@lx01:~$ python test.py
This command is not allowed on the login
Please run resource-intensive commands i
HINT: Use 'srun' to start an interactive
HINT: Use a run node for lightweight int
      See https://docs.sc.hpi.de/cluster
[d.goll@lx01:~$ srun --account=aisc --par
[d.goll@gx15v1:~$ python test.py
Hello World!
[d.goll@gx15v1:~$ exit
exit
```

```
[d.goll@lx01:~$ cat test.sh
#!/bin/bash
#SBATCH --account=aisc
#SBATCH --partition=aisc-batch
#SBATCH --time=00:01:00 # 1 minute
#SBATCH --mem=1G

python test.py
[d.goll@lx01:~$ sbatch test.sh
Submitted batch job 1848135
[d.goll@lx01:~$ ls
bin  job.logs  slurm-1848135.out  test.py  test.sh  workshop-slurm
[d.goll@lx01:~$ cat slurm-1848135.out
Hello World!
```

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Rules and Best Practices

Where to run things:

- rx01/02 are for lightweight tasks only, not a substitute for compute nodes

Resource usage:

- Always specify expected time/memory usage!

Batch jobs:

- Prefer sbatch over interactive jobs for anything serious

Being a good cluster citizen:

- Don’t leave idle interactive sessions running. Release resources you’re not using.

Read the Terms of Usage: https://docs.sc.hpi.de/Terms-of-Usage

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VSCode Setup

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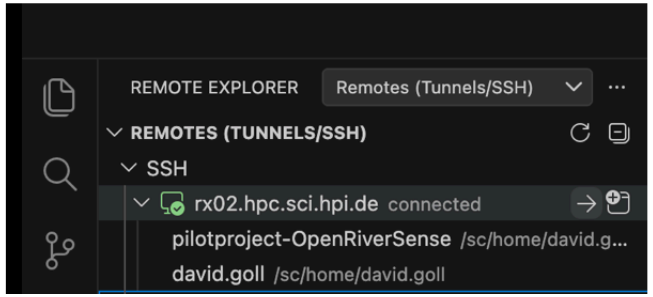
Setting up a Connection with VSCode

When you log in via `ssh user.name@hpc.sci.hpi.de` you are connected with one of the login nodes, e.g. `lx01`. The login nodes are only intended for submitting jobs to Compute nodes, inspect the cluster status, and manage your files. You cannot execute scripts on the login nodes. *They are also not intended for a connection to VSCode.* For this, please use one of the run nodes.

1. In VSCode, open the Command Palette: Shift + Command + P (Mac) / Ctrl + Shift + P (Windows/Linux).
2. Enter `Remote-SSH: Add New SSH Host...`
3. Enter your username and the run node address in the following format: `user.name@rx01.hpc.sci.hpi.de` / `user.name@rx02.hpc.sci.hpi.de`
4. This will write the following information to your `~/.ssh/config` file:

```
Host rx01.hpc.sci.hpi.de
   HostName rx01.hpc.sci.hpi.de
   User user.name
```

5. Now you should be able to connect easily to the run nodes via the Remote Explorer on the left in VSCode:

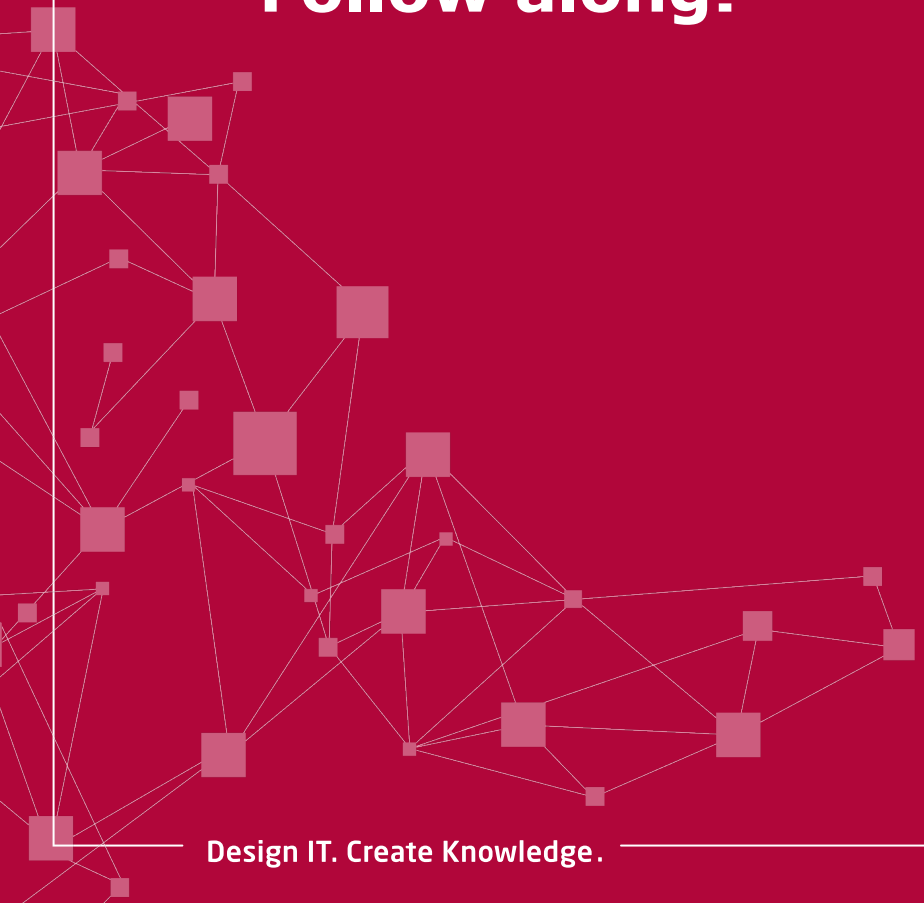




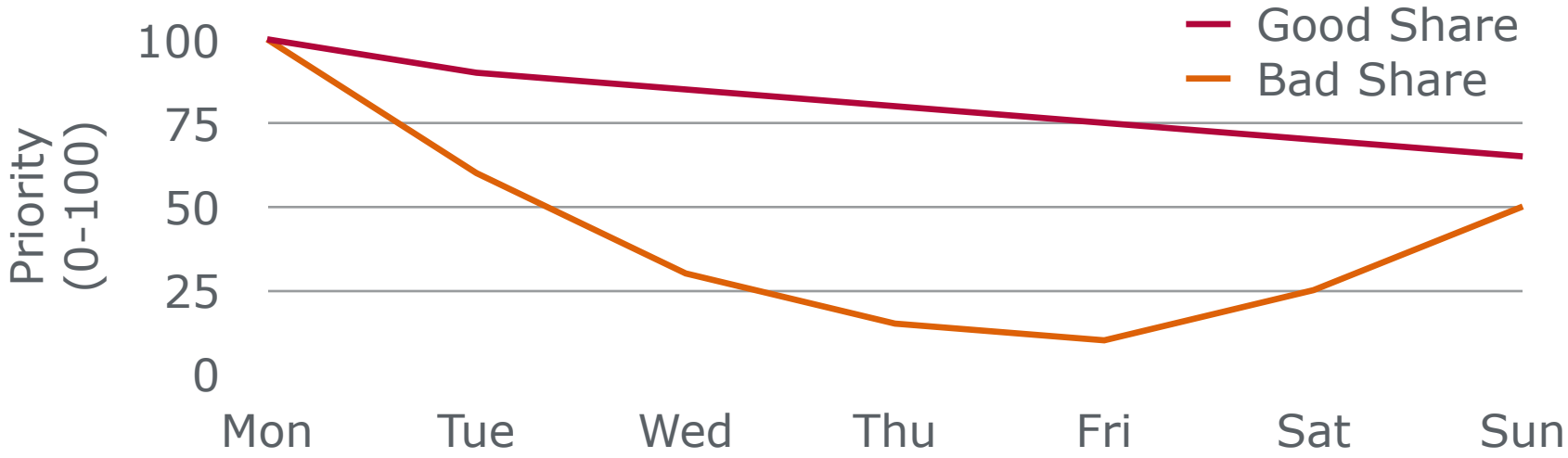
Short Break

Interactive SLURM

Using our Cluster with VSCode interactively. Follow along!



Fairshare



Note: Your priority recovers over time when you're not using resources

- Everyone wants GPUs. SLURM Fairshare decides who goes first.
- Use more → priority drops. Stop using → priority recovers.
- Check your score: `sshare -u $USER`
- Test your pipeline in `pot-hpi-aisc-interactive` first, then submit long runs via `pot-hpi-aisc-batch`.
- **Important:** don't request more than you need, don't leave jobs idle.

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Part II

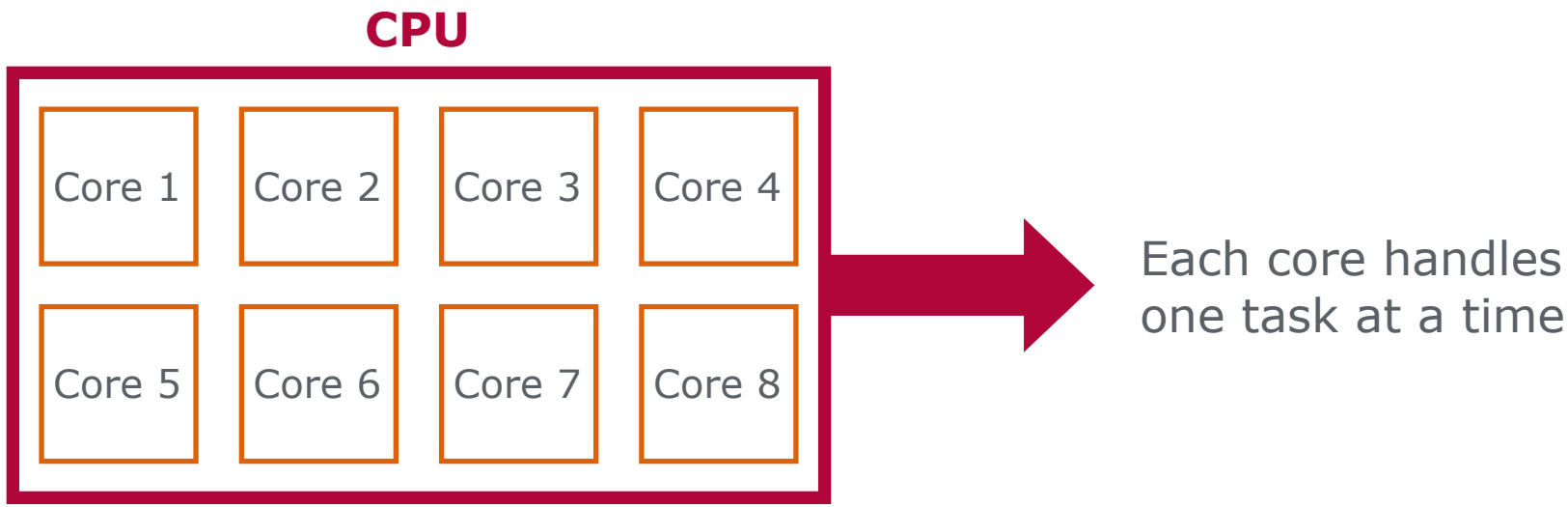
CPUs, GPUs & AI Workloads



CPU

Central Processing Unit

- Few powerful cores, optimized for sequential tasks
- Handles logic, branching, control flow, I/O
- Our H100 nodes:
  - 2x Intel Xeon 8480C, 112 cores (224 threads)



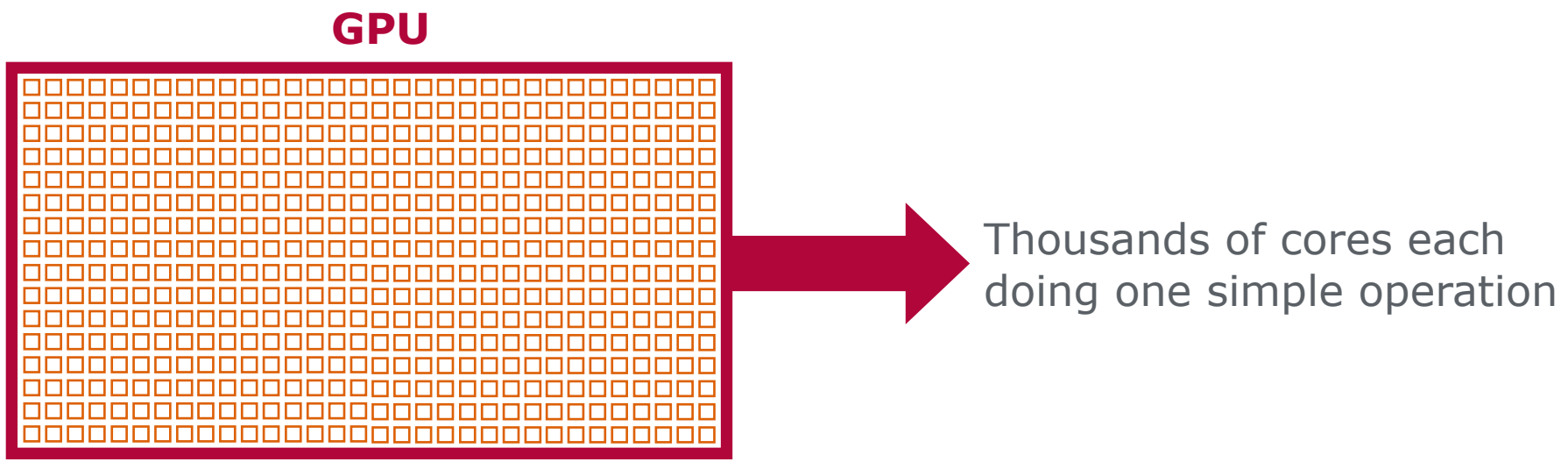
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GPU

Graphics Processing Unit

- Originally designed for rendering pixels on screen
- Thousands of small, simple cores working in parallel
- One H100 has 16,896 CUDA cores (vs 112 CPU cores on the same node)
- Key insight: the same parallelism that renders pixels also powers matrix multiplications in neural networks



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CPU vs GPU

CPU excels at

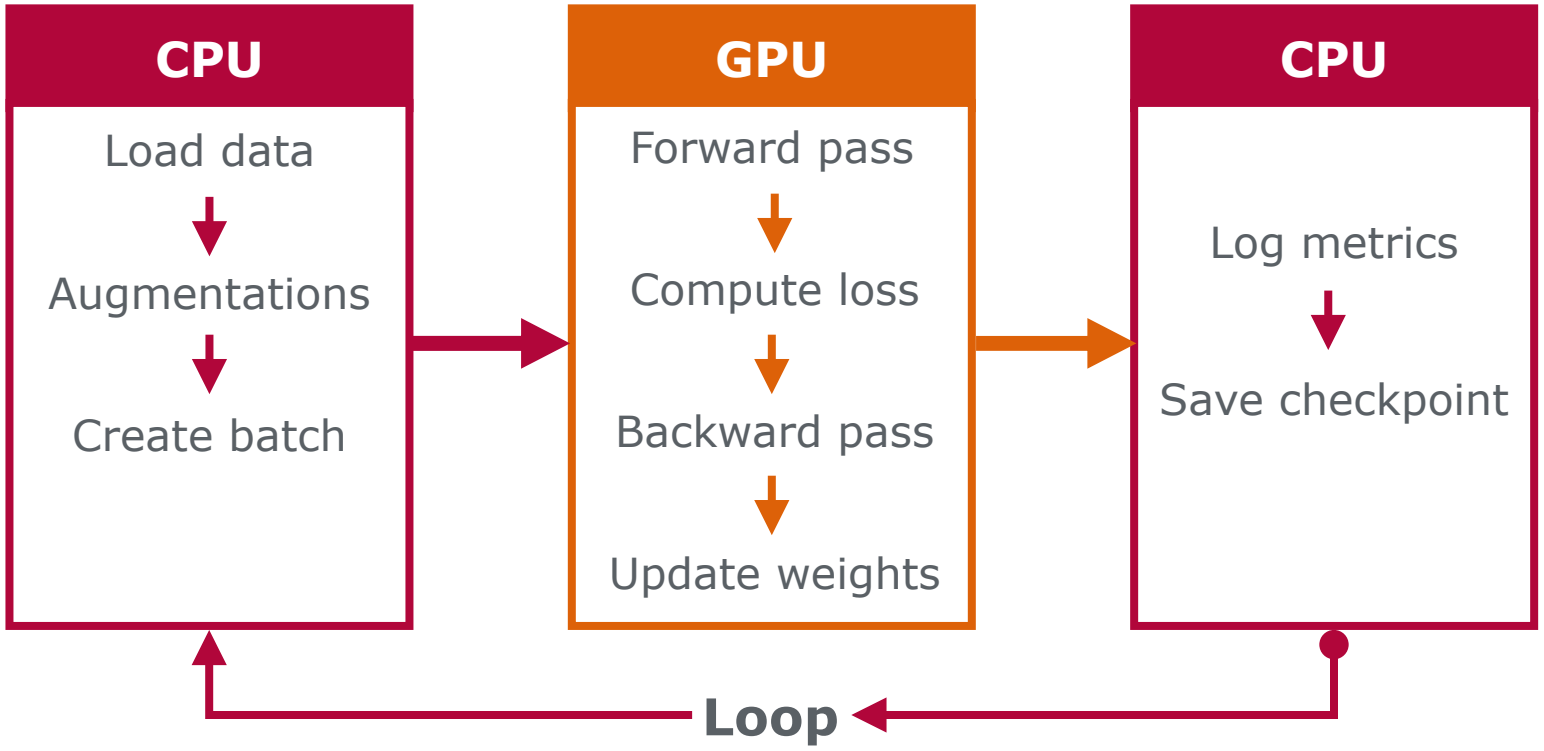
- Data loading and preprocessing (pandas, file I/O)
- Control flow and branching logic
- Classical ML models (sklearn, XGBoost on small data)
- Orchestrating the training pipeline

GPU excels at

- Matrix multiplications (core of every neural network)
- Model Training and Inference
- Applying the same operation to lots of data in parallel (Single Instruction Multiple Data, **SIMD**)

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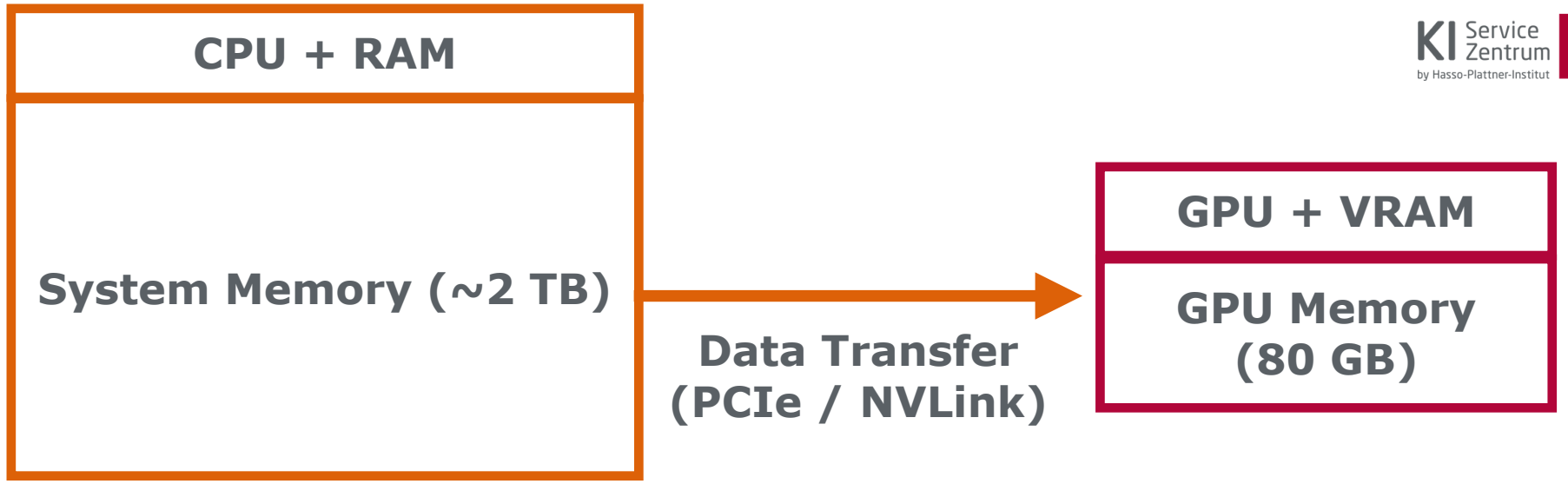
CPU + GPU



- CPU feeds data, GPU does the math
- CPU too slow → GPU sits idle ("CPU bottleneck")
- Request both in SLURM: `--cpus-per-task` and `--gpus`
- Rule of thumb: 4-8 CPUs per GPU
- Our H100 nodes: up to 14 CPU cores per GPU available

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RAM vs VRAM



RAM (system memory):

used by the CPU. Holds your dataset, preprocessing pipelines, OS, and everything else. Our H100 nodes have ~2 TB.

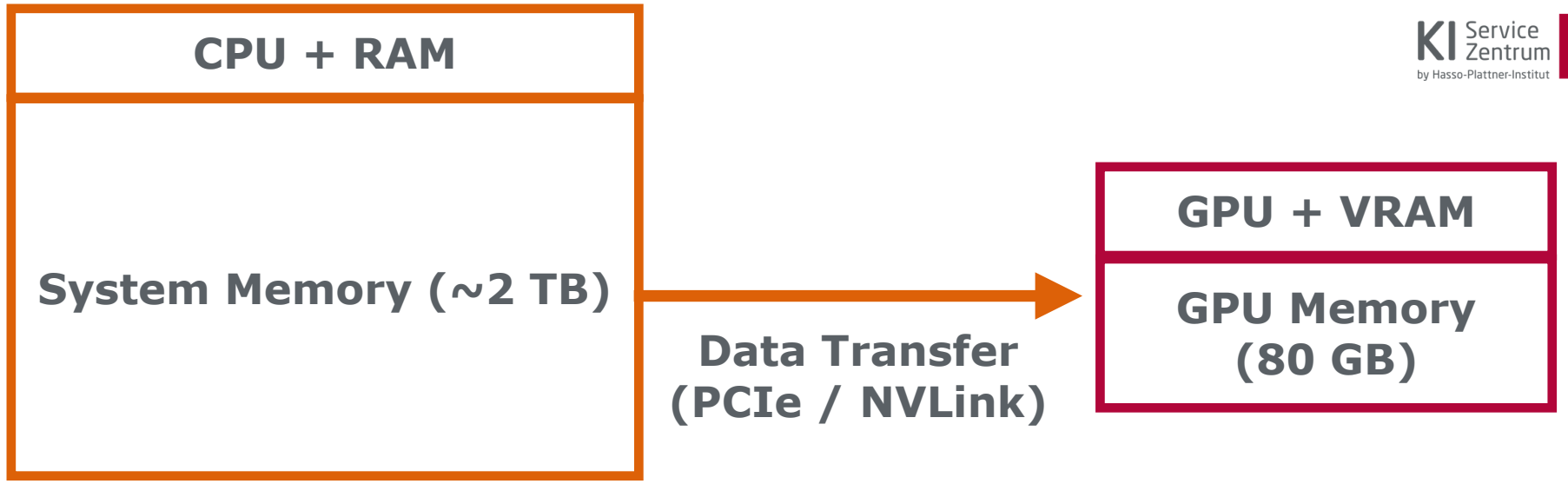
VRAM (GPU memory):

dedicated memory on the GPU. Holds model weights, activations, and gradients during training. An H100 has 80 GB.

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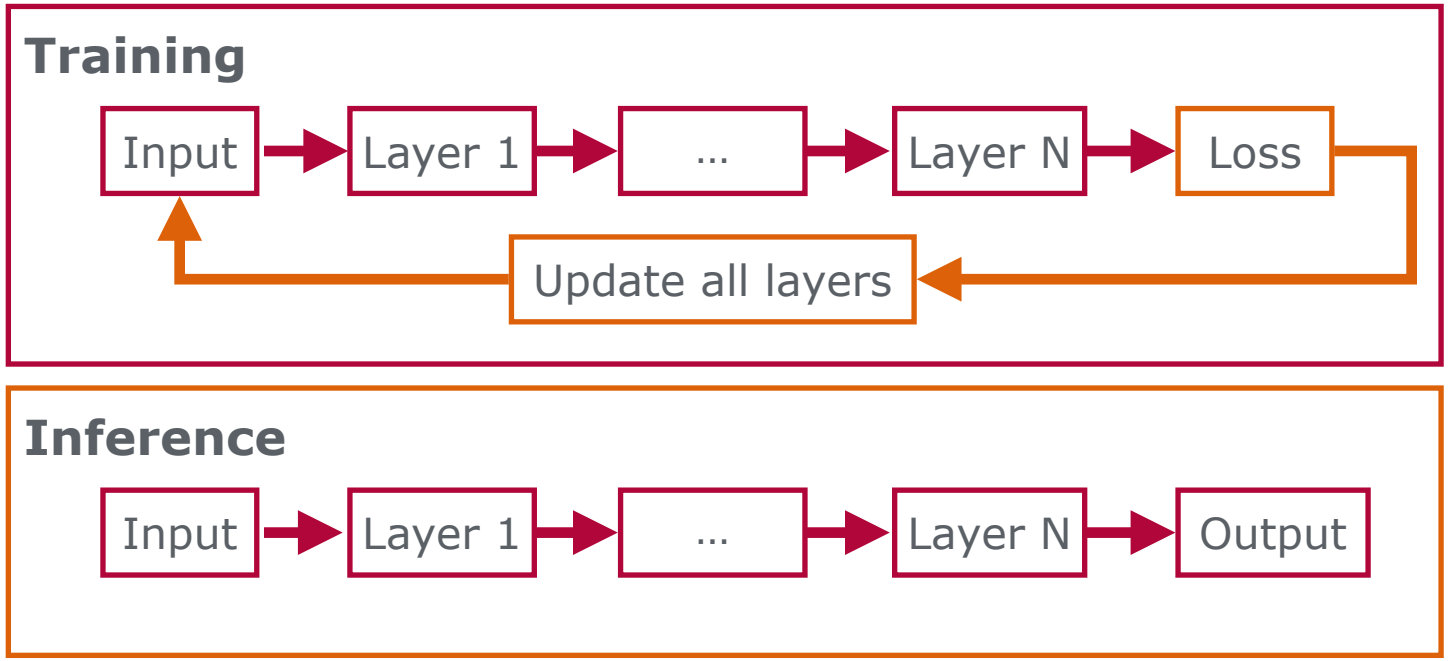
RAM vs VRAM



- 7B parameter model in fp16 ≈ 14 GB just for weights. Add activations and gradients → easily fills 80 GB
- Out of RAM → job gets killed
- Out of VRAM → **CUDA out of memory**

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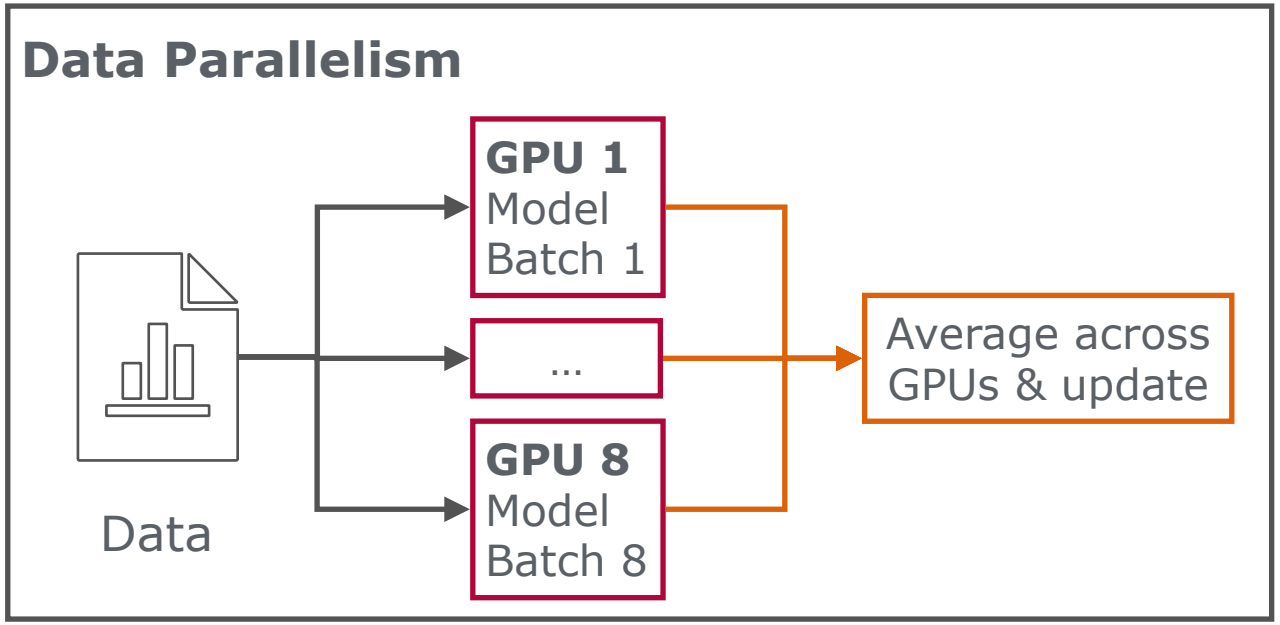
Training vs Inference



- Training: forward + backward pass, store activations, update weights. Repeat millions of times.
- Inference: forward pass only. Less VRAM, less compute.
- H100s (80 GB) for training, A30s (24 GB) for inference.
- Our inference infrastructure runs on Kubernetes with ArgoCD.
- Future workshop planned for model deployment.

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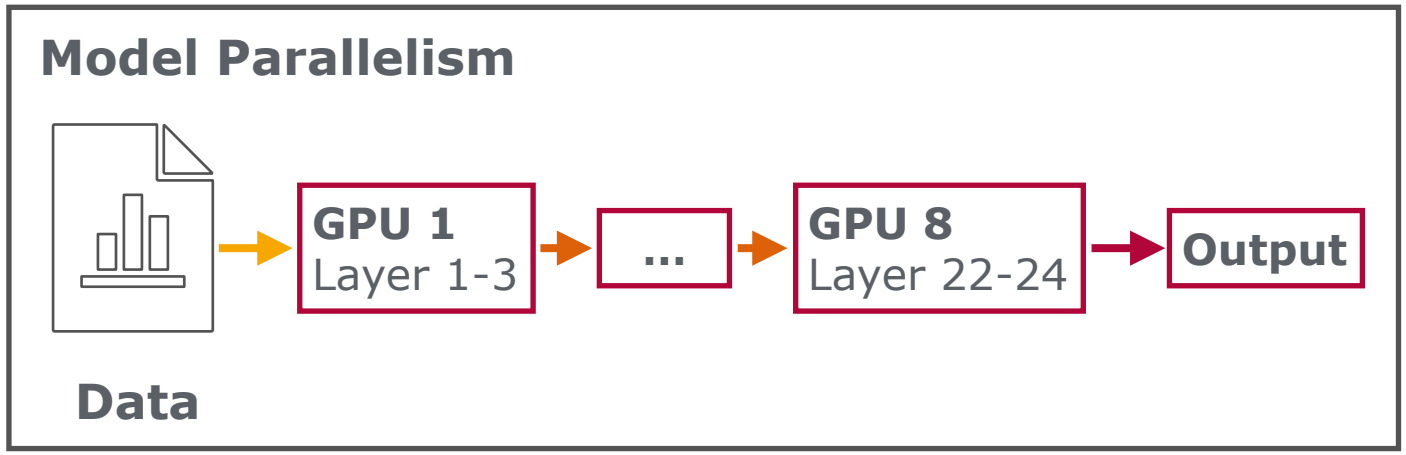
Multi-GPU



- Same model copied to every GPU, each gets a different batch
- GPUs compute in parallel, then sync gradients
- Most common approach, scales training speed roughly linearly
- Frameworks: **torchrun**, **Hugging Face Accelerate**, **DeepSpeed**
- This is what we will use in the hands-on exercise

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Multi-GPU



- Model is split across GPUs, each holds a portion of the layers
- Needed when the model doesn't fit on a single GPU
- Example: 70B LLM in fp16 ≈ 140 GB, doesn't fit on one H100 (80 GB)
- More complex to set up, use when data parallelism isn't enough
- More common in inference, where large models need multiple GPUs just to be loaded
- Start with one GPU, scale up when you hit a bottleneck

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Our GPUs at a Glance

GPU	Count	VRAM	Best for
H100	**64**	**80 GB**	Large model training, multi-GPU jobs
A30	**40**	**24 GB**	Inference, smaller training, interactive queues
L40s	**8**	**48 GB**	Mixed workloads
L40	**2**	**48 GB**	Mixed workloads (ARM server)

Note: `pot-hpi-aisc-interactive` runs on A30s, `pot-hpi-aisc-batch` runs mainly on H100s.

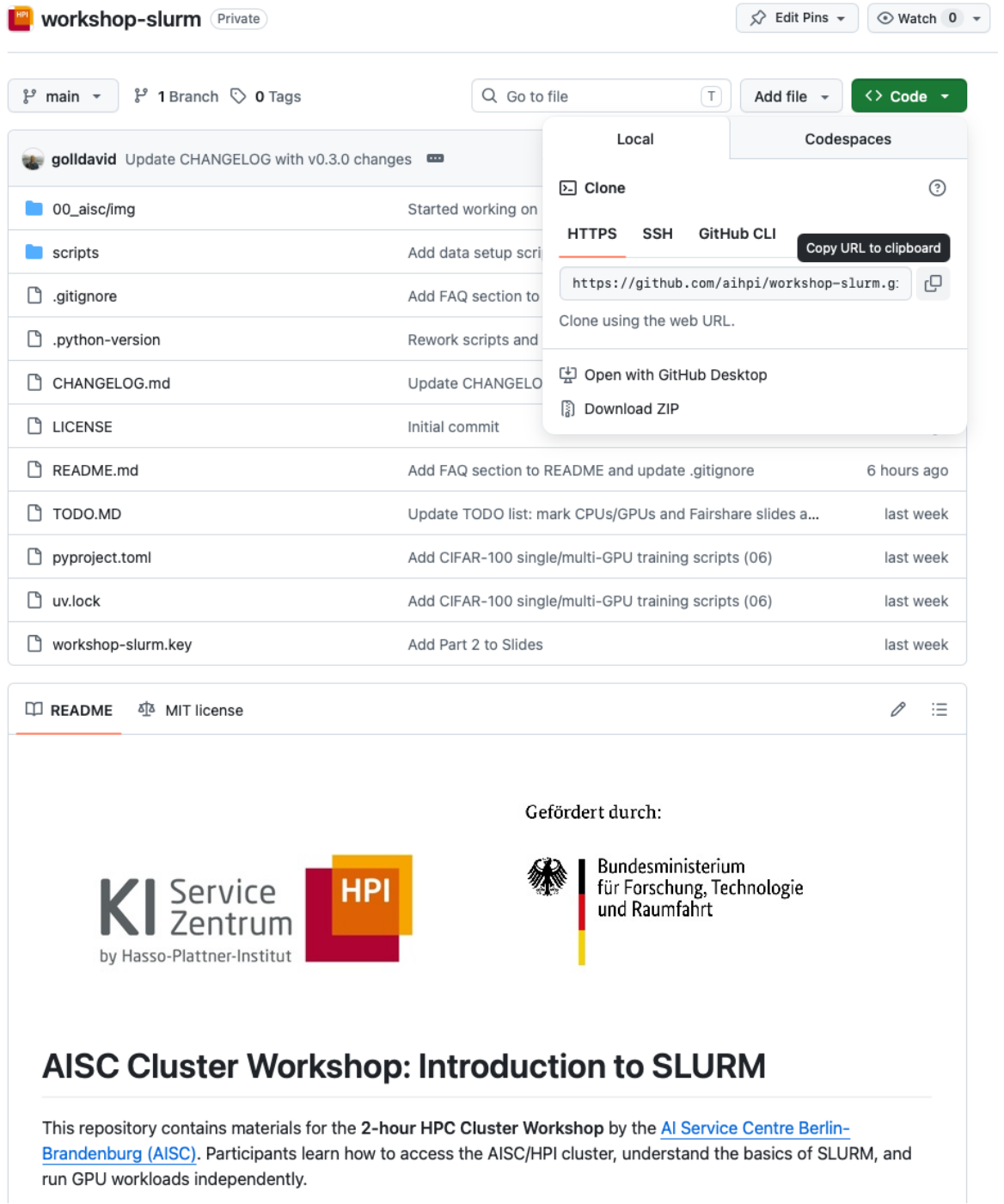
H100 Nodes (gx07-gx14)

- 2x Intel Xeon 8480C, 112 cores / 224 threads
- ~2 TB RAM · 8x H100 80 GB per node · 28 TB NVMe
- NDR InfiniBand + 100G Ethernet
  → 14 CPU cores and ~256 GB RAM per GPU

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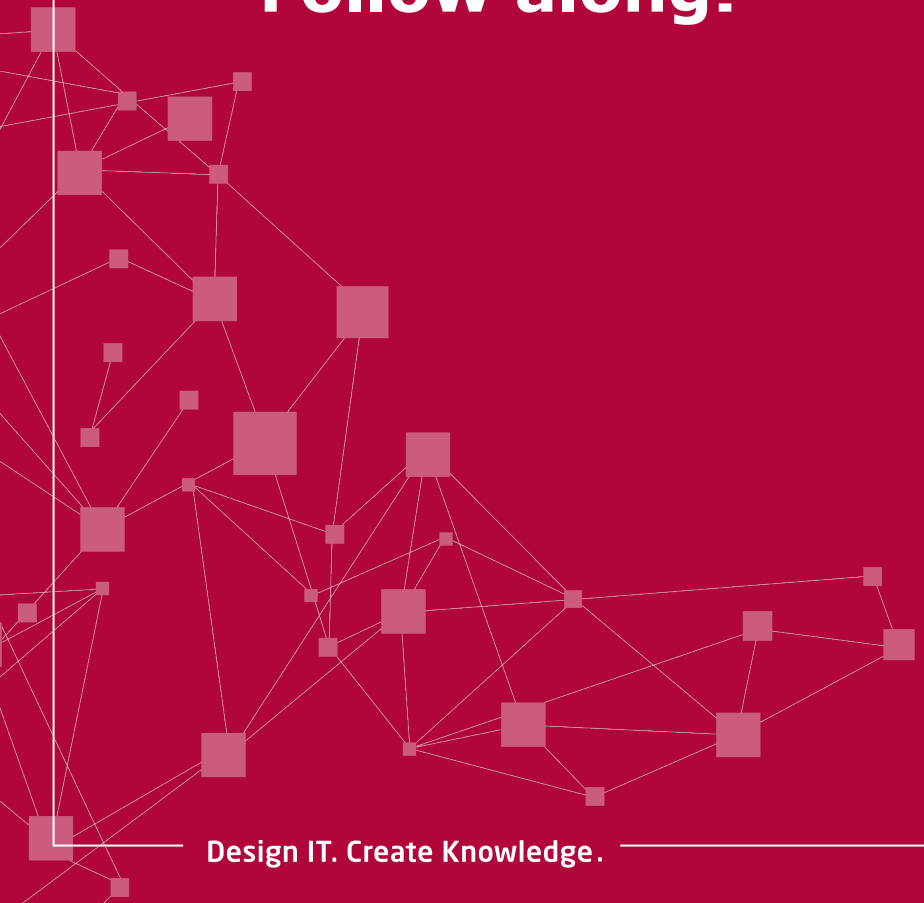
Clone the repository

github.com/aihpi/workshop-slurm



SBATCH Jobs

Submit jobs via scripts.
Follow along!



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Ihre Meinung ist uns wichtig!

QR-Code zum Feedback-Formular

